

DS 2500 *Fall* 2025 Lab Exercise 03

Deadline : 09/22/2025 11:59 PM EST

Late Deadline: 10/06/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Problem 1

```
employees = [  
    {"name": "Alice", "department": "Engineering", "years_experience":  
5},  
    {"name": "Ann", "department": "Marketing", "years_experience": 4},  
    {"name": "Ben", "department": "Engineering", "years_experience":  
2},  
    {"name": "Bob", "department": "Engineering", "years_experience":  
6},  
    {"name": "Eve", "department": "HR", "years_experience": 3},  
]
```

1. Given a list of employee records as dictionaries below, use `list comprehension` to extract the names of employees who:
 1. Work in the "Engineering" department, and
 2. Have more than 3 years of experience
2. Assign the new list to a variable named, `filtered_employees`.

Problem 2

```
pairs = [(5, 2), (1, 4), (3, 1), (2, 9)]
```

1. Given a list of tuples where each tuple contains two integers, use **list comprehension** to create a new list of tuples. In the new list, each tuple should contain three elements where the first two elements are the original integers and the third element is their product.
2. Assign the new list to a variable named, **triples**

Problem 3

```
pairs = [(5, 2), (1, 4), (3, 1), (2, 9)]
```

1. Use the **sorted()** function with a lambda expression to sort the list **pairs** in *ascending order* based on the second element of each tuple.
2. Assign the new list to a variable named, **p3_pairs**.

Problem 4

```
pairs = [(5, 2), (1, 4), (3, 1), (2, 9)]
```

1. Use the **sorted()** function with a lambda expression to sort the list **pairs** in *descending order* based on the sum of each tuple.
2. Assign the new list to a variable named, **p4_pairs**.